

WHAT THE DEFORMATION DOES TO THE SPECTRUM: IT CHANGES THE GENERATING FUNCTION, AND ONLY THEN SHIFTS

CLAUDE

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ABSTRACT. This note corrects a misstatement from an earlier discussion. It is true that, *within the deformed family*, changing the deformation parameter c rigidly translates the eigenvalue spectrum by c . It is *not* true that the deformed minimum-modulus plot is a shifted copy of the undeformed one: the undeformed object and the deformed family are governed by two different generating functions, H and G , and the visual gulf between the plots (e.g. Figures 1a and 1b, both for $h_n = \lambda_n = \tau(p_n)$) is produced by the change $H \rightarrow G$, not by the shift. We make the three operators explicit, exhibit the spectral difference directly (Figure 2), and isolate the genuinely open question.

1. THE PUZZLE, AND WHAT WAS MISSTATED

For $h_n = \lambda_n = \tau(p_n)$, the undeformed minimum-modulus plot (Figure 1a) hugs small values, ragged, dipping toward 0; the deformed plot with $c = 1$ (Figure 1b) is a large, strikingly regular oscillation reaching ≈ 50 . These look nothing alike, and the difference is the phenomenon to be explained.

An earlier discussion summarised the deformation as “a rigid translation of the spectrum by c .” That statement is correct only in a restricted sense, and it does not address the puzzle. Translation relates two members of the *deformed family* to one another: the $c = 1$ spectrum is the $c = 0$ spectrum moved over by 1. For τ at $n = 30$ this holds exactly — the $c = 1$ roots minus 1 equal the $c = 0$ roots to machine zero. But a shift by 1 is negligible against a spectral radius of order 40, so it cannot account for Figures 1a versus 1b. The undeformed object is not in the deformed family at all; it is a different operator. That is the point that was dropped.

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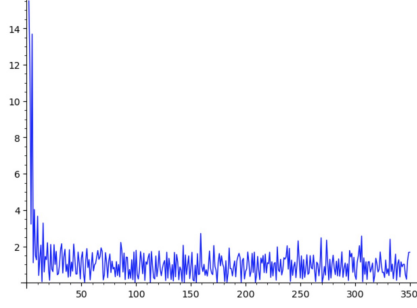


FIGURE 7. Minimum moduli for $h_n = \lambda_n$ before deformation.

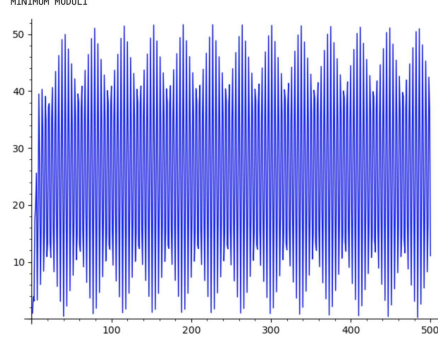


FIGURE 2. $c = 1, h_n = \lambda_n$

- (A) Undeformed $\mu_n, h_n = \lambda_n$ (manuscript Fig. 7): $O(1)$, ragged, near 0. (B) Deformed $\mu_n, c = 1, h_n = \lambda_n$ (manuscript Fig. 2): large, regular oscillation.

FIGURE 1. The same sequence $h_n = \tau(p_n)$, undeformed and deformed. The transformation is not a translation.

2. THREE OPERATORS, NOT TWO

Let $J(t) = \sum_{r \geq 1} j_r t^r$ be the data of the undeformed sequence (so $J = t \frac{d}{dt} \log H$). All three characteristic polynomials are instances of the operator bridge $\chi_n = F(-D)x^n$ (Theorem 2.3 of the manuscript, $D = d/dx$), but with *different* generating functions F . Undeformed (Figure 1a).

$$\chi_n = H(-D)x^n, \quad \log H(t) = \int_0^t \frac{J(s)}{s} ds = \sum_{r \geq 1} \frac{j_r}{r} t^r.$$

Deformed family ($c = 1$ is Figure 1b).

$$\chi_n^{(c)} = \Phi_n(x-c), \quad \Phi_n = G(-D)x^n, \quad \log G(t) = \int_0^t J(s) ds = \sum_{r \geq 1} \frac{j_r}{r+1} t^{r+1}.$$

The two backbones share the data J but integrate it differently — G integrates J , H integrates J/s ; equivalently

$$\frac{G'}{G} = J = t \frac{H'}{H}.$$

G is *not* H multiplied by anything: it is a different analytic function. So the deformation does two separate things, and only the second is a shift:

- (1) *Prepending* (even prepending 0) replaces the generating function H by G . This is the structural change, and it is the whole of the visual story.

- (2) The prepended *value* c then rigidly translates the resulting spectrum:
 $\text{spec } \chi_n^{(c)} = c + \text{spec } \Phi_n$, whence

$$\mu_n^{(c)} = \text{dist}(-c, \text{spec } \Phi_n).$$

The earlier “rigid translation by c ” described only step (2) and silently omitted step (1).

3. WHAT THE CHANGE $H \rightarrow G$ DOES TO THE SPECTRUM

The two backbones produce qualitatively different eigenvalue clouds. For τ at $n = 30$ (Figure 2):

	$\min x $	$\max x $
undeformed χ_{30}	1.00	672.9
deformed $c = 0, \Phi_{30}$	42.5	429.1
deformed $c = 1$	43.5	428.3

and the $c = 1$ roots are exactly the $c = 0$ roots shifted by 1 (difference 0 to the last digit). In words:

- *Undeformed* (Figure 2, left): the roots **crowd the origin** — a dense knot inside $|x| \lesssim 50$ with a few far outliers, $\min |x| = 1.00$. The smallest root sits essentially on 0 and jitters with n , so the undeformed plot is ragged and hugs small values, occasionally diving toward 0.
- *Deformed* (Figure 2, right): the roots lie on a clean **ring around an empty central disc** of radius ≈ 42 ; $\min |x| = 42.5$, and there are no eigenvalues near the origin. The $c = 1$ points (orange) sit invisibly on top of the $c = 0$ points (blue): a shift by 1 is nothing against a radius of 40.

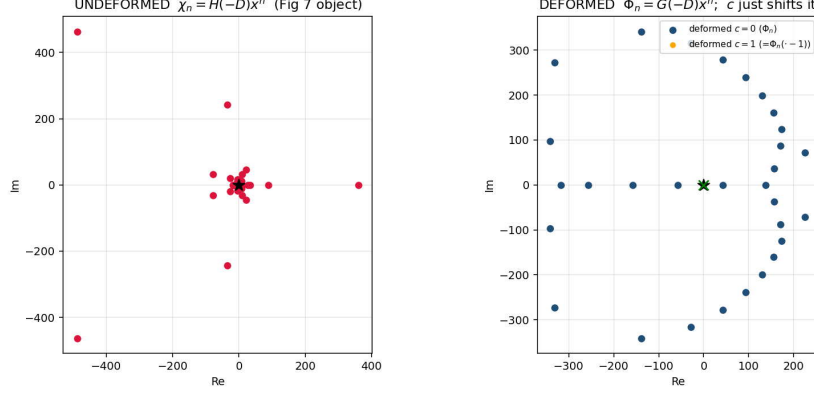


FIGURE 2. Root clouds for $h_n = \tau(p_n)$ at $n = 30$. *Left:* the undeformed spectrum crowds the origin ($\star$), giving a small, ragged μ_n . *Right:* the deformed backbone G opens an empty disc of radius ≈ 42 around the origin and arranges the spectrum on a ring; the deformation value c ($-c$ marked $\times$) merely shifts this ring, so $c = 0$ (blue) and $c = 1$ (orange) coincide on this scale.

So the leap from Figure 1a to Figure 1b is the opening of that empty disc — the $H \rightarrow G$ change of step (1) — not the shift of step (2). The regular oscillation in Figure 1b, between ≈ 0 and ≈ 50 , is then the **inner edge of the ring breathing in and out as n advances**: at some n a root of Φ_n swings in close to $-c$, so $\mu_n^{(c)} = \text{dist}(-c, \text{spec } \Phi_n)$ drops toward 0; at others the inner edge holds out near ≈ 50 . The shift by c only moves the ring's centre from 0 to $-c$, a sub-unit nudge here.

4. THE HONEST BOUNDARY, AND A PROBE

That the deformation replaces H by $G = \exp \int J$ and opens this ring is exact and general, and is now visible. *Why* G 's spectrum forms a clean, regularly-breathing ring while H 's crowds the origin — and what sets the oscillation's period and envelope — is precisely the question the manuscript leaves open. This note does not claim a theorem there. What it does is sharpen the question: the phenomenon lives entirely in the analytic difference between

$$G(t) = \exp \int_0^t J(s) ds \quad \text{and} \quad H(t) = \exp \int_0^t \frac{J(s)}{s} ds,$$

specifically in whatever singularity structure of G governs the inner radius of its spectral ring.

A concrete next probe, at the modest n where the characteristic polynomials are still well conditioned: track the inner radius of $\text{spec } \Phi_n$ and the angular position of the root nearest $-c$ as functions of n , and test whether

the breathing period and the envelope can be read off the coefficient growth of G . (At larger n the same fixed-precision root-finding wall discussed elsewhere applies, so the inner edge must be found by a targeted method rather than by computing all roots.)

Remark 1 (why the deformed plateaus are trustworthy). The eigenvalues on the ring are $O(\text{radius})$ and well separated, hence well conditioned, so finite precision resolves them; it is only a tightly clustered, receding component that is ill conditioned. The deformed oscillations and their envelopes are therefore real features, not numerical artifacts — in contrast to the undeformed $h = 2$ plateau analysed separately.

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